

Data Analyst vs Data Scientist vs AI Engineer

Steps involved in the **AI/ML Pipeline**

- Step1 : Data Collection
- Step2 : Data Cleaning & Preprocessing
- Step3 : Exploratory Data Analysis (EDA)
- Step4 : Feature Engineering
- Step5 : Model Selection & Training
- Step6 : Model Evaluation
- Step7 : Model Deployment

Skills for different Career Paths

1. Data Analyst

- Data Collection
 - SQL
 - Excel / Google Sheets
- Data Cleaning & Preprocessing
 - Python
 - Statistics
- EDA
 - Power BI
 - Tableau

2. Data Scientist

- Data Collection
 - SQL

- Data Cleaning & Preprocessing
 - Advanced Python
 - NumPy & Pandas
- EDA
 - Matplotlib
 - Seaborn
- Feature Engineering
- Model Selection & Training
 - Scikit-learn
 - Supervised ML
 - Unsupervised ML
 - Reinforcement Learning
 - Git & Github (additional)
 - Math
 - Statistics
 - Linear Algebra
 - Probability
 - Calculus
- Model Evaluation

3. AI Engineer

- Data Collection
 - SQL
- Data Cleaning & Preprocessing
 - Advanced Python
 - NumPy & Pandas
- EDA

- Matplotlib
 - Seaborn
- Feature Engineering
- Model Selection & Training
 - Scikit-learn
 - Supervised ML
 - Unsupervised ML
 - Reinforcement Learning
 - Git & Github (additional)
 - Math
 - Statistics
 - Linear Algebra
 - Probability
 - Calculus
 - Deep Learning (PyTorch or TensorFlow)
 - ANN
 - FNN
 - RNN
 - CNN
 - Transformer etc.
- Model Evaluation
- Model Deployment
 - Basic Deployment
 - API Building
 - DSA (optional, asked by some companies)